

4 PRINTED CIRCUIT BOARDS

The printed circuit boards are factory-set for operation. Usually, you do not need to set or adjust them. This chapter describes the standard settings and adjustment required if a defective printed circuit board is replaced. It also describes the test pins and the LED indications.

The controller printed circuit board includes the main unit printed circuit board and one or more cards or modules installed horizontally to the main-unit printed-circuit board.

These PC boards have interface connectors, LED indicators, and a plastic panel at the front. At the rear, there is a backplane connector.

4.1 MAIN BOARD

Card and Modules

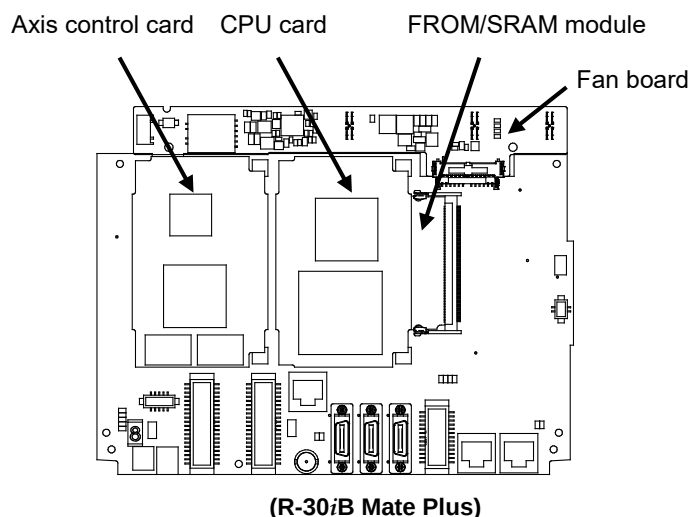
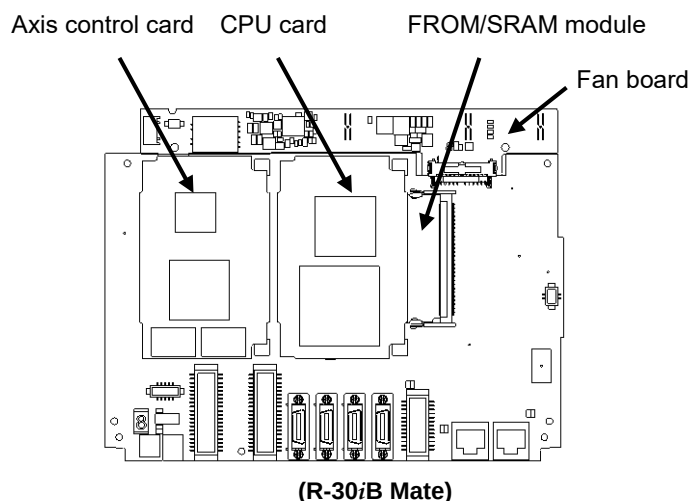


Fig.4.1 Main board

Name	Ordering Specification	Board Specification	Note
Main board	A05B-2650-H001	A20B-8200-0790	Standard, Ethernet:1ch
	A05B-2650-H002	A20B-8200-0791	Ethernet:2ch, Vision I/F, Force sensor I/F
	A05B-2650-H003	A20B-8200-0792	Ethernet:2ch, Vision I/F, Force sensor I/F, PMC, HDI
	A05B-2650-H004	A20B-8201-0420	Standard, Ethernet:1ch For I/O Link <i>i</i> slave
	A05B-2650-H005	A20B-8201-0421	Ethernet:2ch, Vision I/F, Force sensor I/F For I/O Link <i>i</i> slave
	A05B-2650-H006	A20B-8201-0422	Ethernet:2ch, Vision I/F, Force sensor I/F, PMC, HDI For I/O Link <i>i</i> slave
	A05B-2680-H001 (R-30iB Mate Plus)	A20B-8201-0750	Standard, Ethernet:2ch For I/O Link <i>i</i> slave
	A05B-2680-H002 (R-30iB Mate Plus)	A20B-8201-0751	Ethernet:3ch, Vision I/F, Force sensor I/F For I/O Link <i>i</i> slave
	A05B-2680-H003 (R-30iB Mate Plus)	A20B-8201-0752	Ethernet:3ch, Vision I/F, Force sensor I/F, PMC, HDI For I/O Link <i>i</i> slave
CPU card	A05B-2600-H020	A20B-3300-0686	Standard / SDRAM 32Mbyte
		A17B-3301-0106	
	A05B-2600-H021	A20B-3300-0687	Standard / SDRAM 64Mbyte
		A17B-3301-0107	
	A05B-2600-H022	A20B-3300-0688	Standard / SDRAM 128Mbyte
		A17B-3301-0108	
	A05B-2600-H023	A20B-3300-0683	High speed / SDRAM 32Mbyte
		A17B-3301-0103	
	A05B-2600-H024	A20B-3300-0684	High speed / SDRAM 64Mbyte
		A17B-3301-0104	
	A05B-2600-H025	A20B-3300-0685	High speed / SDRAM 128Mbyte
		A17B-3301-0105	
	A05B-2600-H026	A17B-3301-0109	Standard / SDRAM 32Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H027	A17B-3301-0110	Standard / SDRAM 64Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H028	A17B-3301-0111	Standard / SDRAM 128Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H029	A17B-3301-0112	High speed / SDRAM 32Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H030	A17B-3301-0113	High speed / SDRAM 64Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H031	A17B-3301-0114	High speed / SDRAM 128Mbyte For I/O Link <i>i</i> slave
	A05B-2670-H020 (R-30iB Mate Plus)	A17B-3301-0250	Standard / DRAM 1GB For I/O Link <i>i</i> slave

Name	Ordering Specification	Board Specification	Note
Axis control card	A05B-2600-H040	A20B-3300-0664	6-axis
		A20B-3300-0774	
	A05B-2600-H041	A20B-3300-0663	12-axis
		A20B-3300-0773	
	A05B-2600-H042	A20B-3300-0662	18-axis
		A20B-3300-0772	
	A05B-2600-H043	A20B-3300-0661	24-axis
		A20B-3300-0771	
	A05B-2600-H044	A20B-3300-0660	36-axis
		A20B-3300-0770	
	A05B-2670-H040 (R-30iB Mate Plus)	A20B-3300-0819	6-axis
	A05B-2670-H041 (R-30iB Mate Plus)	A20B-3300-0818	12-axis
	A05B-2670-H042 (R-30iB Mate Plus)	A20B-3300-0817	18-axis
	A05B-2670-H043 (R-30iB Mate Plus)	A20B-3300-0816	24-axis
	A05B-2670-H044 (R-30iB Mate Plus)	A20B-3300-0815	36-axis
FROM/SRAM module	A05B-2600-H060	A20B-3900-0283	FROM 32M/ SRAM 1M
		A20B-3900-0297	
	A05B-2600-H061	A20B-3900-0284	FROM 32M/ SRAM 2M
		A20B-3900-0298	
	A05B-2600-H062	A20B-3900-0285	FROM 32M/ SRAM 3M
		A20B-3900-0299	
	A05B-2600-H063	A20B-3900-0286	FROM 64M/ SRAM 1M
	A05B-2600-H064	A20B-3900-0287	FROM 64M/ SRAM 2M
	A05B-2600-H065	A20B-3900-0288	FROM 64M/ SRAM 3M
	A05B-2600-H066	A20B-3900-0280	FROM 128M/ SRAM 1M
	A05B-2600-H067	A20B-3900-0281	FROM 128M/ SRAM 2M
	A05B-2600-H068	A20B-3900-0282	FROM 128M/ SRAM 3M
	A05B-2600-H069 (R-30iB Mate Plus)	A20B-3900-0293	FROM 256M/ SRAM 1M
	A05B-2600-H070 (R-30iB Mate Plus)	A20B-3900-0295	FROM 256M/ SRAM 2M
	A05B-2600-H071 (R-30iB Mate Plus)	A20B-3900-0296	FROM 256M/ SRAM 3M
Fan board	A05B-2650-H001	A20B-8200-0669	
	A05B-2650-H002		
	A05B-2650-H003		

NOTE

In case of using the function of I/O Link *i* slave, the combination of the specification of mainboard, CPU card and software is limited as below.

[Main board (For I/O Link *i* slave)] + [CPU card (For I/O Link *i* slave)]
+ [Software (V8.30P14 or later)]

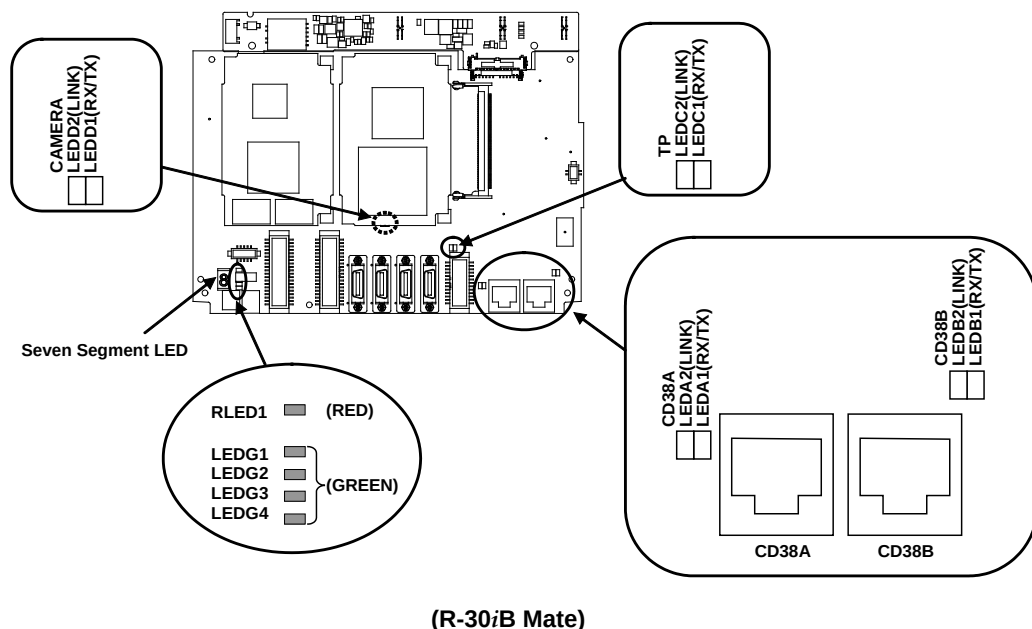
Except for the above condition, the system does not work correctly as followings.

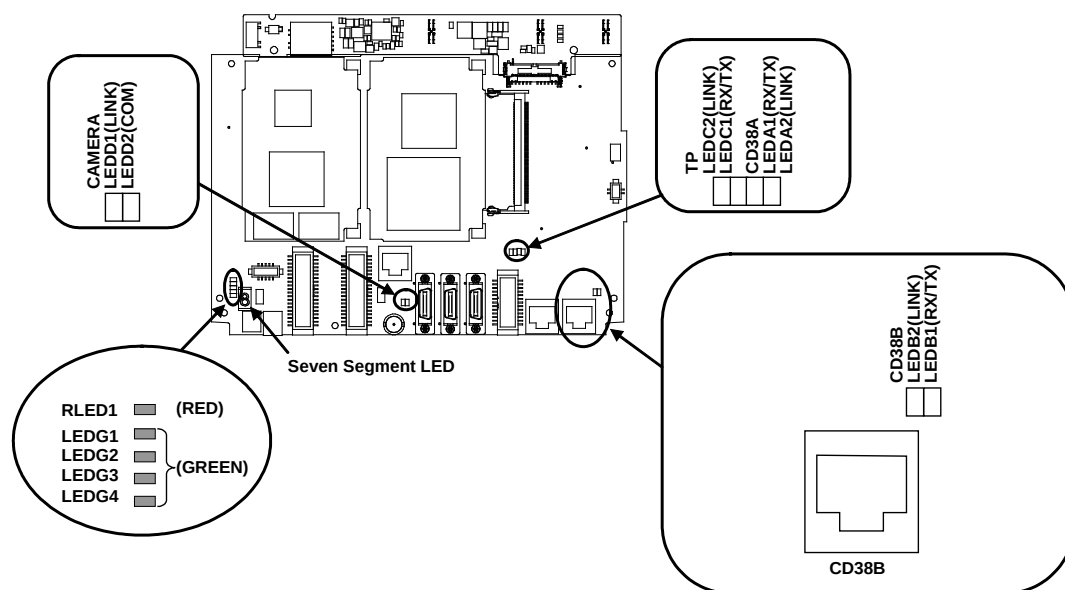
- a) [Main board] + [CPU card (For I/O Link *i* slave)]
Neither the function of I/O Link slave nor I/O Link *i* slave work correctly. And the alarm regarding I/O Link or I/O Link *i* occur on the master side.
And in case the software version is not correct, system does not work correctly.
- b) [Main board (For I/O Link *i* slave)] + [CPU card]
The system does not work correctly.

In case of using the R-30iB Mate Plus, the combination of the specification of mainboard, CPU card and software is limited as below.

[Main board (R-30iB Mate Plus)] + [CPU card (R-30iB Mate Plus)]
+ [Software (V9.00P03 or later)]

Except for the above condition, the system does not work correctly. And CPU card (R-30iB Mate Plus) may be broken.

LEDs



(R-30iB Mate Plus)

Seven segment LED	Description
	When the alarm condition has occurred in the main board, this LED is turned on. Please see the Section 3.7.TROUBLESHOOTING BASED ON LED INDICATIONS.

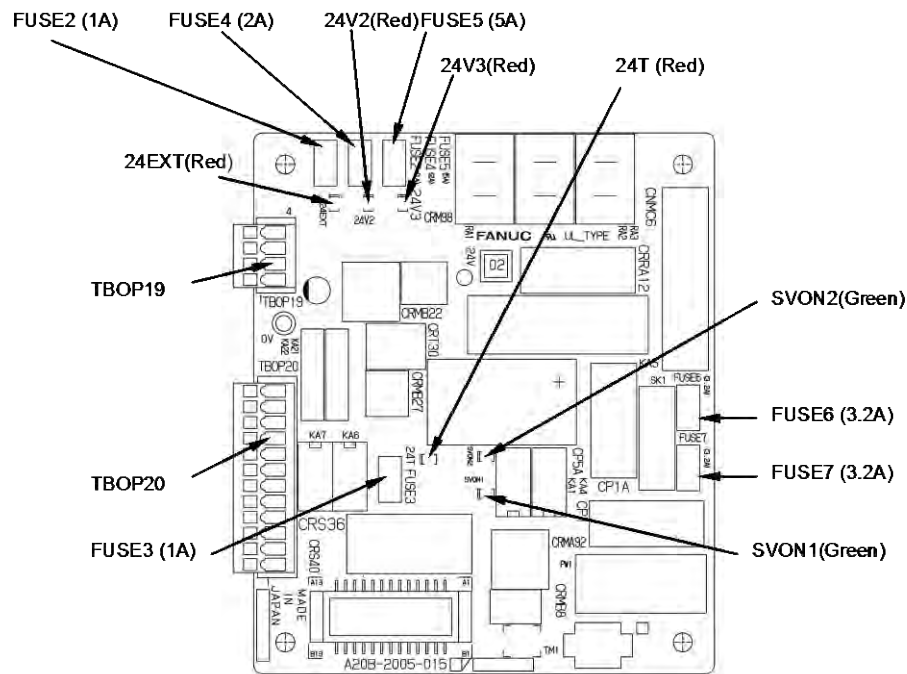
ALARM LED	Color	Description
RLED1	Red	When the alarm condition has occurred in the CPU card, this LED is turned on. Please see the Section 3.7.TROUBLESHOOTING BASED ON LED INDICATIONS.

STATUS LED	Color	Description
LEDG1	Green	These LEDs show the operating status of the system. Please see the Section 3.7.TROUBLESHOOTING BASED ON LED INDICATIONS.
LEDG2	Green	
LEDG3	Green	
LEDG4	Green	

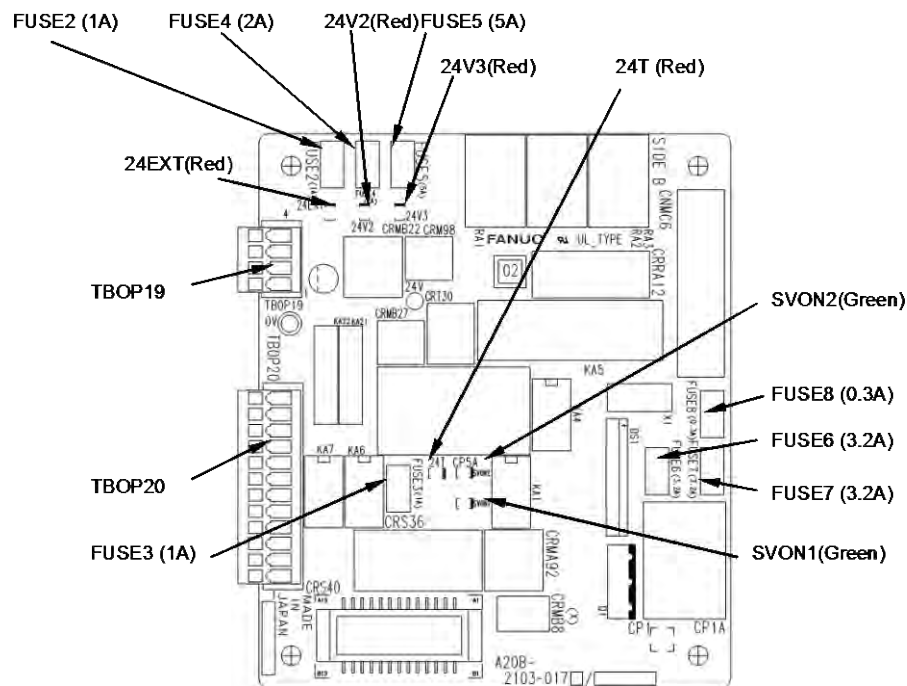
ETHERNET LED	Color	Description
RX/TX	Green	Blink during data transmission of Ethernet TP
LINK	Green	Light when a link of Ethernet is established

Note. TP:For Ethernet TP, CAMERA: For camera

4.2 EMERGENCY STOP BOARD (A20B-2005-0150, A20B-2103-0170)



(R-30i/B Mate A20B-2005-0150)



(R-30i/B Mate Plus A20B-2103-0170)

Fig.4.2 Emergency stop board

4.3 BACKPLANE

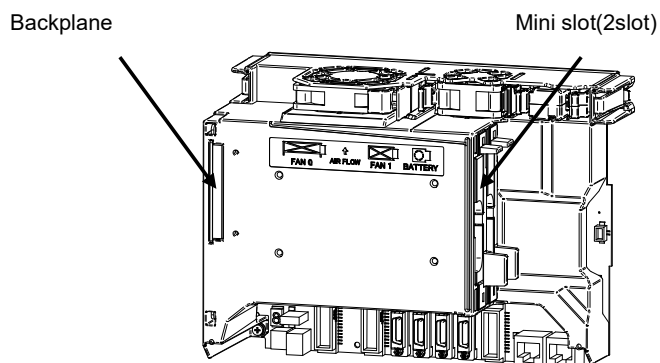


Fig.4.3 Backplane

Name	Ordering Specification	Parts number	Board specification
2 slot backplane	A05B-2650-H080	A05B-2650-C040	A20B-8200-0680

4.4 PROCESS I/O BOARD MA (A20B-2004-0381)

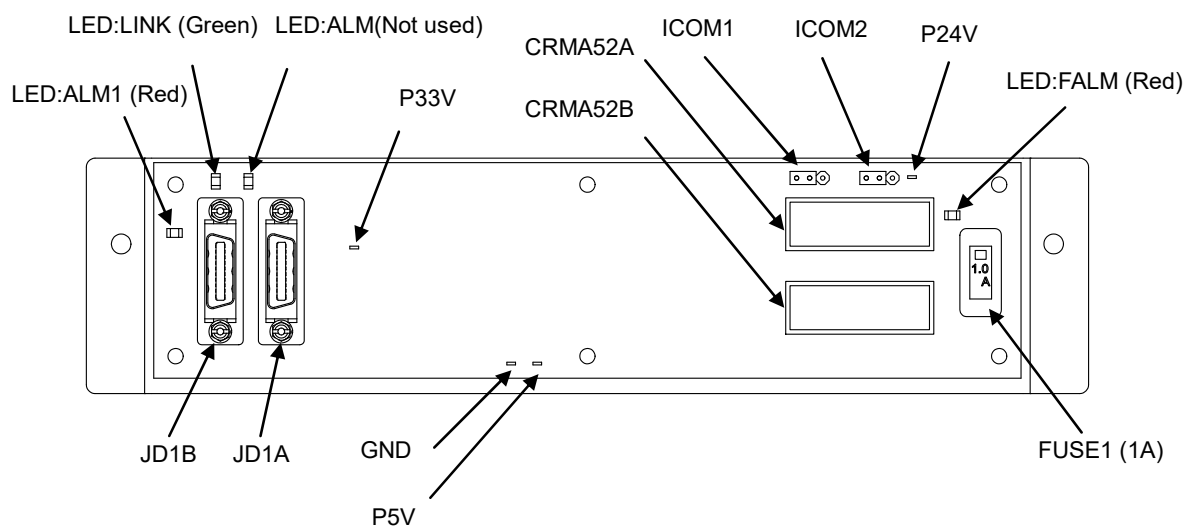


Fig.4.4 Process I/O Board MA

(1) Test pins

Name		Use
P24V	+24V	For measuring the DC supply voltage
P5V	+5V	
GND	GND	
P33V	+3.3V	

(2) Settings

Name		Standard setting	Description
ICOM1	UDI1- 10 (Connector CRMA52A)	Side A	For common voltage setting Side A: +24V common Side B: 0V common
ICOM2	UDI11- 20 (Connector CRMA52B)		

(3) LEDs

Name	Color	Description
ALM1	Red	A communication alarm occurred between the main board and process I/O board.
FALM	Red	The fuse (FUSE1) on the process I/O board has blown.
LINK	Green	Blink (1:3) Communication in progress ON = approx. 0.5 sec, OFF = approx. 1.5 sec
		Blink (1:1) Communication not in progress ON = approx. 0.25 sec, OFF = approx. 0.25 sec
ALM	Red	Not used

4.5 PROCESS I/O BOARD MB (A20B-2101-0731)

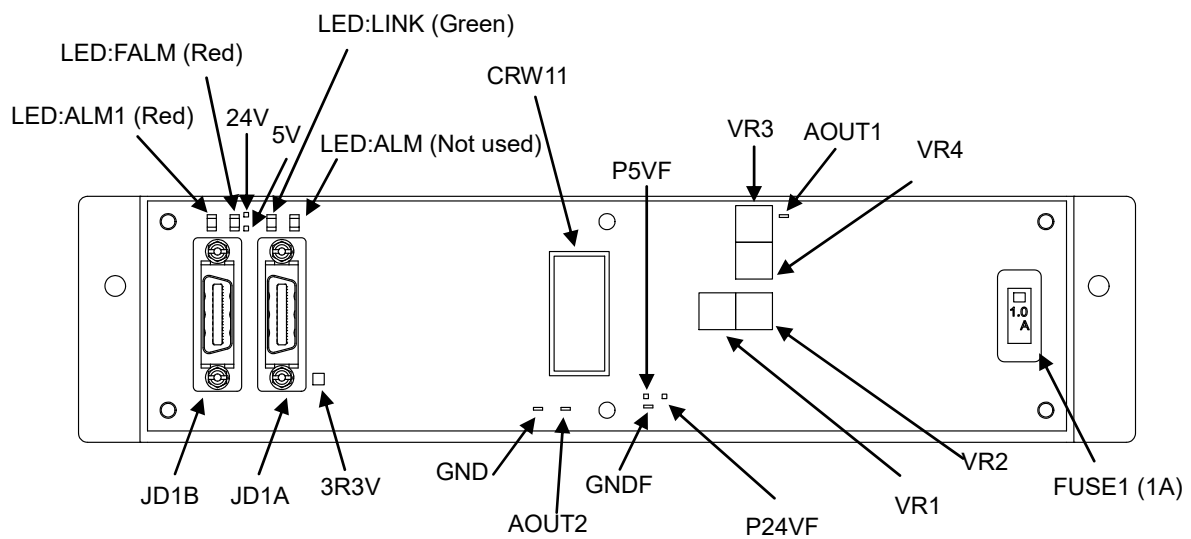


Fig.4.5 Process I/O Board MB

(1) Test pins and pads

Name		Use
24V	+24V	For measuring the DC supply voltage
5V	+5V	
GND	GND	
3R3V	+3.3V	
P24VF	+24V	D/A converter power supply
P5VF	+5V	
GNDF	GND	
AOUT1	Channel 1	For analog output signal (D/A) voltage measurement
AOUT2	Channel 2	

(2) Adjustment

VR1/VR2 Channel 1 gain and offset adjustment

Connect the “+” and “-” terminals of a digital voltmeter, respectively, to the AOUT1 check pin and the GNDF check pin (not a general ground point). From the teach pendant, execute AOUT[1]=3413, using a robot program. While observing the voltage at the AOUT1 check pin with the digital voltmeter, adjust potentiometers VR1 and VR2 for 15.0V.

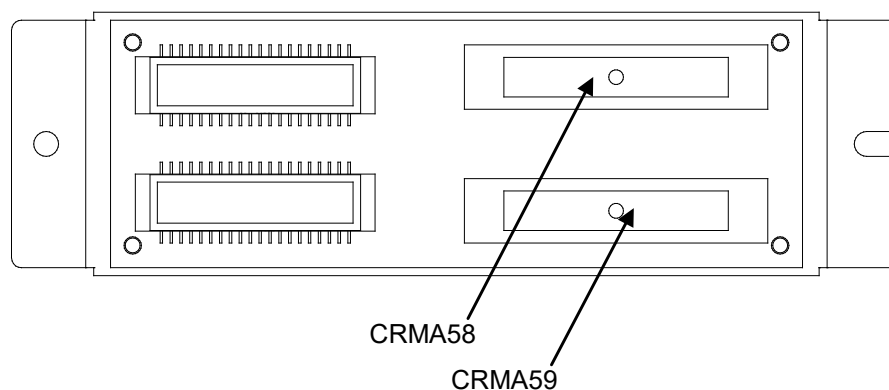
VR3/VR4 Channel 2 gain and offset adjustment

Connect the “+” and “-” terminals of a digital voltmeter, respectively, to the AOUT2 check pin and the GNDF check pin (not a general ground point). From the teach pendant, execute AOUT[2]=3413, using a robot program. While observing the voltage at the AOUT2 check pin with the digital voltmeter, adjust potentiometers VR3 and VR4 for 15.0V.

(3) LEDs

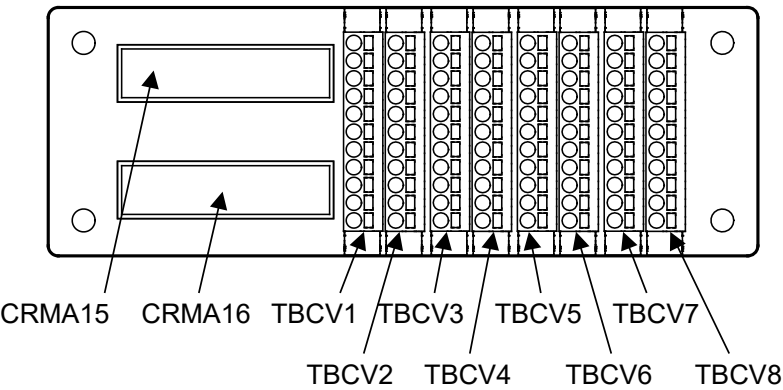
Name	Color	Description
ALM1	Red	A communication alarm occurred between the main CPU and process I/O board.
FALM	Red	The fuse (FUSE1) on the process I/O board has blown.
LINK	Green	Blink (1:3) Communication in progress ON = approx. 0.5 sec, OFF = approx. 1.5 sec Blink (1:1) Communication not in progress ON = approx. 0.25 sec, OFF = approx. 0.25 sec
ALM	Red	Not used

4.6 CONNECTOR CONVERTER BOARD (A20B-2004-0411)



(Connector converter board)

4.7 TERMINAL CONVERTER BOARD (A20B-1009-0690)



(Terminal converter board)

5

6-AXIS SERVO AMPLIFIERS

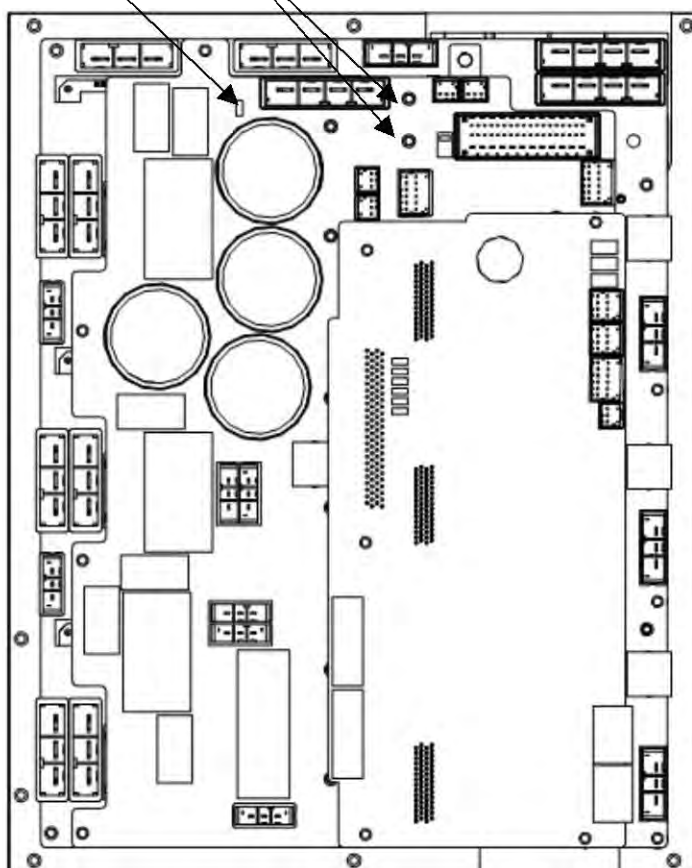
The 6-axis servo amplifiers are factory-set for operation. Usually, you do not need to set or adjust them. This chapter describes the standard settings and adjustment required if a defective 6-axis servo amplifier is replaced. It also describes the use of test pins and meanings of the LED indications.

Table 5 6-axis servo amplifier specification (Power supply regeneration)

ROBOT	6-AXIS SERVO AMPLIFIER	REGENERATIVE RESISTOR
ARC Mate 120iC, M-20iA, M-20iB M-710iC, R-1000iA	A06B-6400-H102	A05B-2650-C101
R-2000iC, M-2iA, M-3iA	A06B-6400-H002	A05B-2650-C101
ARC Mate 100iC M-10iA	A06B-6400-H003	A05B-2650-C101
ARC Mate 50iD	A06B-6400-H005	A05B-2650-C101
LR Mate 200iD, M-1iA ARC Mate 50iD (Limited type) CR-4iA, CR-7iA	A06B-6400-H005	A05B-2650-C100

Check that the voltage is not higher than 50V.

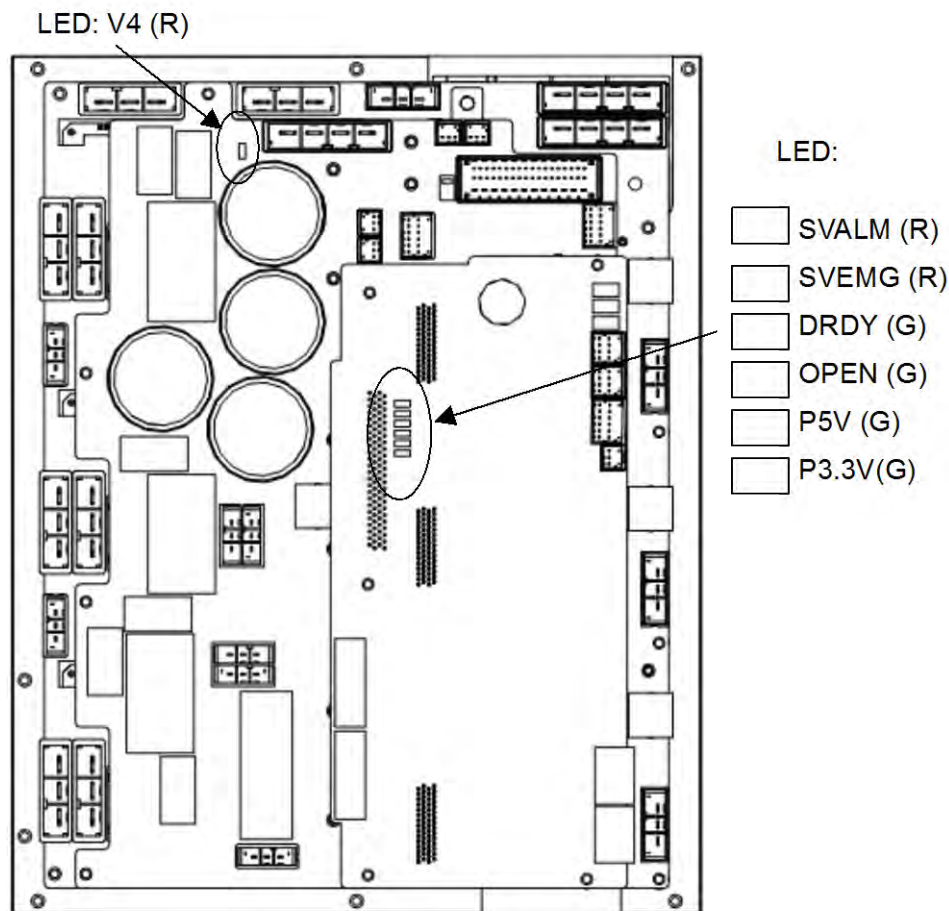
LED : V4 (red)



WARNING

Before touching the servo amplifier, for example, for maintenance purposes, check the voltage at the screw above the LED "V4" with a DC voltage tester to see if the remaining voltage is not higher than 50V.

5.1 LEDS OF 6-AXIS SERVO AMPLIFIER



LED	Color	Description
V4	Red	Lights when the DCLINK circuit inside the servo amplifier is charged to reach a specific voltage.
SVALM	Red	Lights when the servo amplifier detects an alarm.
SVEMG	Red	Lights when an emergency stop signal is input to the servo amplifier.
DRDY	Green	Lights when the servo amplifier is ready to drive the servo motor.
OPEN	Green	Lights when the communication between the servo amplifier and the main board is normal.
P5V	Green	Lights when the power supply circuit inside the servo amplifier outputs a voltage of +5 V normally.
P3.3V	Green	Lights when the power supply circuit inside the servo amplifier outputs a voltage of +3.3 V normally.

5.2 SETTING OF 6-AXIS SERVO AMPLIFIER

Table 5.2 6-axis servo settings

Name	Standard setting	Description
COM1	Side A	Robot Digital Input (RI) device common voltage. Side A: +24V common Side B: 0V common

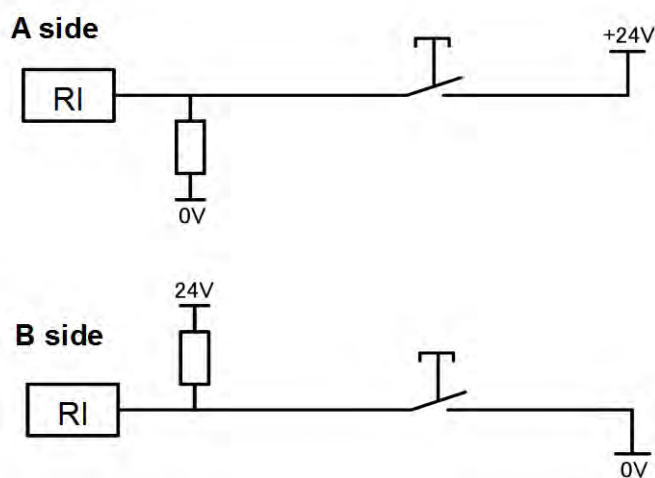
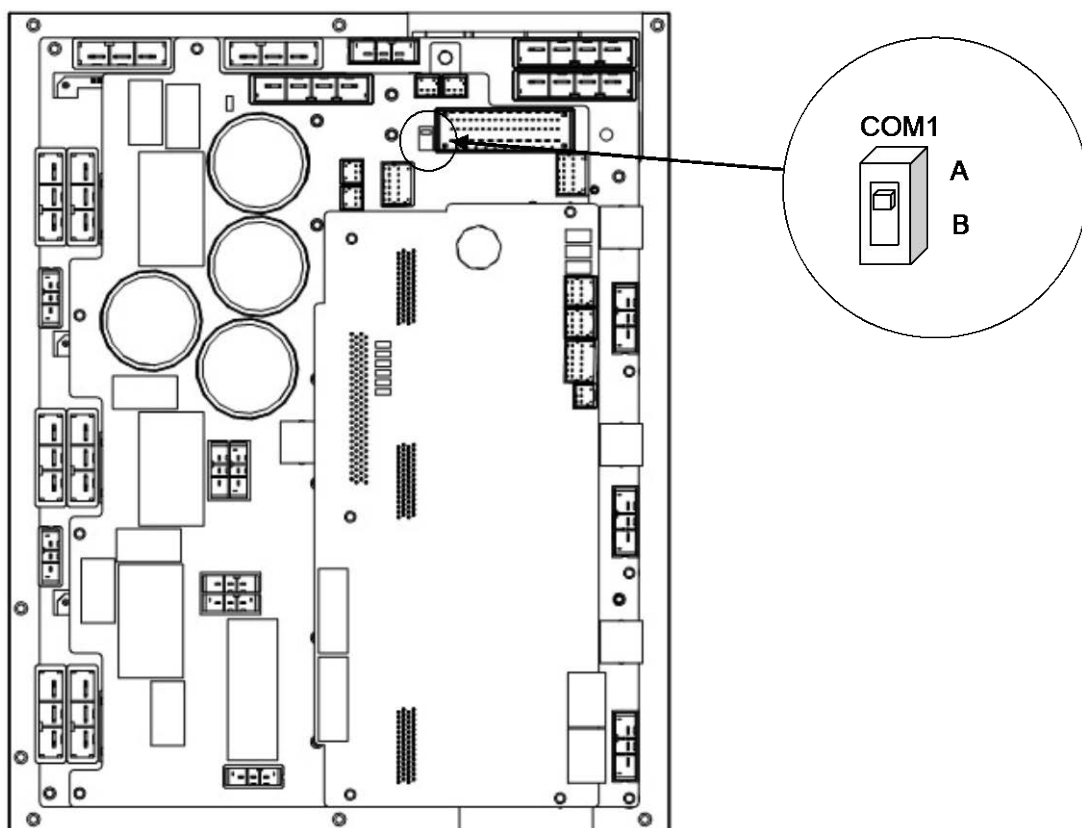


Fig.5.2 Circuit based on jumper pin location or setting of switch

5.3 6-AXIS SERVO AMPLIFIER SPECIFICATIONS

SPECIFICATIONS TABLE : (A06B-6400-H***)

UNIT		A06B-6400-H102	A06B-6400-H002	A06B-6400-H003
INPUT RATINGS	VOLTAGE	AC200~AC240V (+10% / -15%), 50/60Hz, 3phase		
	POWER CAPACITY	5.1KVA	5.1KVA	2.7KVA
OUTPUT RATINGS	MAXIMUM OUTPUT	240V ~		
	CURRENT : J1	80Ap / 23.0Arms	80Ap / 23.0Arms	40Ap / 13.4Arms
	CURRENT : J2	80Ap / 23.0Arms	80Ap / 23.0Arms	40Ap / 13.4Arms
	CURRENT : J3	80Ap / 23.0Arms	80Ap / 23.0Arms	20Ap / 6.5Arms
	CURRENT : J4	40Ap / 13.4Arms	40Ap / 13.4Arms	20Ap / 6.5Arms
	CURRENT : J5	40Ap / 13.4Arms	40Ap / 13.4Arms	20Ap / 6.5Arms
	CURRENT : J6	40Ap / 13.4Arms	40Ap / 13.4Arms	20Ap / 6.5Arms
	TOTAL CURRENT	70Arms	70Arms	30Arms

UNIT		A06B-6400-H005
INPUT RATINGS	VOLTAGE	AC200~AC240V (+10% / -15%), 50/60Hz, 3/1phase
	POWER CAPACITY	1.3/1.4 (3/1phase)
OUTPUT RATINGS	MAXIMUM OUTPUT	240V ~
	CURRENT : J1	20Ap / 3.6Arms
	CURRENT : J2	20Ap / 3.6Arms
	CURRENT : J3	20Ap / 3.6Arms
	CURRENT : J4	20Ap / 3.6Arms
	CURRENT : J5	10Ap / 2.0Arms
	CURRENT : J6	10Ap / 2.0Arms
	TOTAL CURRENT	18.4Arms